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Chengalpattu District 2026

Class : 12

Register Number

COMMON QUARTERLY EXAMINATION - 2026-27

CHEMISTRY

Time Allowed : 3.00 Hours] [Max. Marks : 70

PART – I

I. **Choose the correct answer.** 15x1=15

1. Considering Ellingham diagram, which of the following metals can be used to reduce alumina?
   a) Fe b) Cu c) Mg d) Zn
2. Which of these is not a monomer for a high molecular mass silicone polymer?
   a) Me_3SiCl b) $PhSiCl_3$ c) $MeSiCl_3$ d) Me_2SiCl_2
3. Among the following, which is the strongest oxidising agent?
   a) Cl_2 b) F_2 c) Br_2 d) I_2
4. The number of moles of acidified $KMnO_4$ required to oxidize 1 mole of ferrous oxalate (FeC_2O_4) is
   a) 5 b) 3 c) 0.6 d) 1.5
5. The fraction of total volume occupied by the atoms in a simple cubic is
   a) $\left(\frac{\pi}{4\sqrt{2}}\right)$ b) $\left(\frac{\pi}{6}\right)$ c) $\left(\frac{\pi}{4}\right)$ d) $\left(\frac{\pi}{3\sqrt{2}}\right)$
6. After 2 hours, a radio active element substance becomes $\left(\frac{1}{16}\right)^{th}$ of Original amount. Then the half life (in min) is
   a) 60 minutes b) 120 minutes c) 30 minutes d) 15 minutes
7. The P^H of an aqueous solution is zero. The solution is
   a) Slightly acidic b) Strongly acidic c) neutral d) basic
8. **Assertion** : Phenol is more acidic than ethanol
   Reason : Phenoxide ion is reasonance stabilised
   a) both assertion and reason are true and reason is the correct explanation of assertion
   b) both assertion and reason are true but reason is not the correct explanation of assertion
   c) assertion is true and reason is false d) both assertion and reason are false
9. The formation of cyanohydrin from acetone is an example of
   a) nucleophilic substitution b) electrophilic substitution
   c) electrophilic addition d) Nucleophilic addition.
10. Which of the following reduces Tollen's reagent
    a) Formic acid b) acetic acid c) benzophenone d) none of these
11. Glycerol undergoes dehydration to form
    a) Propene b) Prop 2-enal c) 2-Propanone d) 2-butene
12. Which of the following compounds can acts as Lewis acid
    a) NH_3 b) MgO c) SF_4 d) PF_3
13. The hybridisation is $XeOF_2$ is
    a) sp^3 b) sp^3d c) Sp^3d^2 d) dsp^2
14. The rate constant of a reaction is $5.8x10^{-2}\ s^{-1}$. The order of the reaction is
    a) First order b) zero order c) Second order d) Third order
15. Which of the following is not sp^2 hybridised?
    a) Fullerence b) Graphite c) Graphene d) Diamond

PART - II

II. **Answer any Six questions of the following. Question No.24 is compulsory.** 6x2=12

16. Which type of Ores can be concentrated by Froth flotation method? Give two examples for such Ores.

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17. Give the uses of alum?
18. What is inert pair effect?
19. Write the chromyl chloride test?
20. Classify the following solids.
    a) Urea b) Silicon Carbide c) Brass d) Sodium Chloride
21. What is an Order of the reaction.
22. Define P^H.
23. Why C-O-C bond angle in ether is slightly greater than the tetrahedral bond angle.
24. Identify X and Y

$$CH_3COCH_2CH_2COOC_2H_5 \xrightarrow{CH_3MgBr} X \xrightarrow{H_3O^+} Y$$

PART - III

III. Answer any six questions of the following. Question No.33 is compulsory. **6x3=18**

25. Give the limitations of Ellingham diagram?
26. What is catenation? Discuss briefly the catenation property of Carbon.
27. Give the balanced equation for the reaction between chlorine with cold NaOH and hot NaOH.
28. Which is stronger reducing agent Cr^{2+} or Fe^{2+}?
29. Distinguish between hexagonal close packing and Cubic close packing.
30. Define solubility product. Write the expression for the solubility product of Hg_2Cl_2.
31. How is phenol prepared from
    (i) Chlorobenzene (ii) isopropyl benzene
32. What happens when formaldehyde reacts with ammonia? Give its structure.
33. The decomposition of Cl_2O_7 at 500K in the gas phase to Cl_2 and O_2 is a first order reaction. After 1 minute at 500K, the pressure of Cl_2O_7 falls from 0.08 to 0.04 atm. Calculate the rate constant in s^{-1}?

PART - IV

IV. Answer all the questions. **5x5=25**

34. (a) i) Explain the principle of Electrolytic refining with an example? **(3)**
    ii) Describe a method for refining Nickel? **(2)**

    (OR)

    (b) Describe the structure of Diborane. **(5)**
35. (a) i) State any three properties of interhalogen compounds. **(3)**
    ii) HF is not stored in glass bottles. Why? **(2)**

    (OR)

    (b) Compare Lanthanoids and Actinoids. **(5)**
36. (a) Write a note on non-stoichiometric defects. **(5)**

    (OR)

    (b) Derive integrated rate law for a first order reaction. A $\rightarrow$ Products. **(5)**
37. (a) i) Discuss the Lewis concept of acids and bases. **(3)**
    ii) Define Ionic product of water. Give its value at room temperature. **(2)**

    (OR)

    (b) Explain Lucas test. **(5)**
38. (a) **Write a note on :** (i) Haloform reaction **(3)** (ii) Benzoin condensation **(2)**

    (OR)

    (b) Write the mechanism of Aldol condensation. **(5)**

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